

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date:	30 September 2026
Team ID:	SWTID-2026-1128
Project Name:	Auto Ticket Classification using Flow Designer
Maximum Marks:	4 Marks

1. Technical Architecture Overview

The Auto Ticket Classification System leverages ServiceNow's cloud infrastructure, Flow Designer automation engine, and Natural Language Understanding (NLU) / Predict Intelligence components. The architecture is organized across three primary tiers: Client Interface, Platform Orchestration Logic, and Cloud Data/Storage Engine.

- **1. User Request Ingestion:** End-users (Students & Teachers) log incidents via the Service Portal UI or REST API integration.
- **2. NLU & Text Parsing:** ServiceNow NLU engine parses the short description, extracts intent entity keywords, and computes confidence scores for categories.
- **3. Flow Designer Trigger:** Flow Designer executes predefined workflow rules to evaluate conditions, set Category/Subcategory fields, and calculate initial priority.
- **4. Data Store Query & Queue Assignment:** The platform references Assignment Group tables to automatically assign the ticket to the designated IT Support Group.
- **5. Notification Dispatch:** Notification Engine sends automated alerts (Email/SMS) to callers and assigns technicians instantly.

2. Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Portal interface for students & teachers to submit tickets and track status.	ServiceNow Service Portal, HTML5, CSS3, AngularJS, Bootstrap
2	Application Logic-1 (Workflow)	Automated orchestration, trigger handling, and routing workflow.	ServiceNow Flow Designer, JavaScript (ES6 / GlideScript)
3	Application Logic-2 (Text Mining)	Intent classification, keyword parsing, and confidence scoring.	ServiceNow Natural Language Understanding (NLU) Engine
4	Application Logic-3 (Notifications)	Automated email notifications and caller status dispatch logic.	ServiceNow Notification Engine, Notification Triggers
5	Database	Relational database storing incident records, user profiles, and assignment groups.	ServiceNow Relational Database (MariaDB enterprise cluster)

S.No	Component	Description	Technology
6	Cloud Database	Cloud-hosted database service managing multi-tenant data storage.	ServiceNow High Availability Cloud Data Center Storage
7	File Storage	Storage for ticket attachments (log files, screenshots, diagnostic error reports).	ServiceNow Cloud Attachment Storage (AES-256 Encrypted)
8	External API-1	REST API for integrating third-party IT tools and monitoring alerts.	REST API / JSON Web Services
9	External API-2	User identity authentication API against School Active Directory/LDAP.	OAuth 2.0 / Single Sign-On (SSO) / Azure AD Integration
10	Machine Learning Model	Supervised classification model for predicting ticket categories based on historic resolution data.	Predictive Intelligence / Text Categorization Model
11	Infrastructure (Server / Cloud)	Cloud deployment infrastructure ensuring high availability and fault tolerance.	ServiceNow Multi-Instance Enterprise Cloud Architecture

3. Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open web libraries and front-end frameworks used in user interface development.	AngularJS, Bootstrap 4, jQuery, FontAwesome
2	Security Implementations	Comprehensive role-based access control, encrypted communications, and OAuth authentication.	TLS 1.3, AES-256 Encryption, Role-Based Access Control (RBAC), OAuth 2.0, OWASP Standards
3	Scalable Architecture	Multi-tier microservices-enabled platform architecture designed to scale seamlessly with ticket volume.	ServiceNow Multi-Instance Cloud Architecture & Load Balanced Application Tier
4	Availability	High-availability clustering with automatic failover across redundant data centers.	Active-Active Multi-datacenter Redundancy & Load Balancing
5	Performance	Sub-second response time for classification workflows, supported by caching and asynchronous flow execution.	GlideRecord Query Caching, Asynchronous Event Processing, CDN Assets